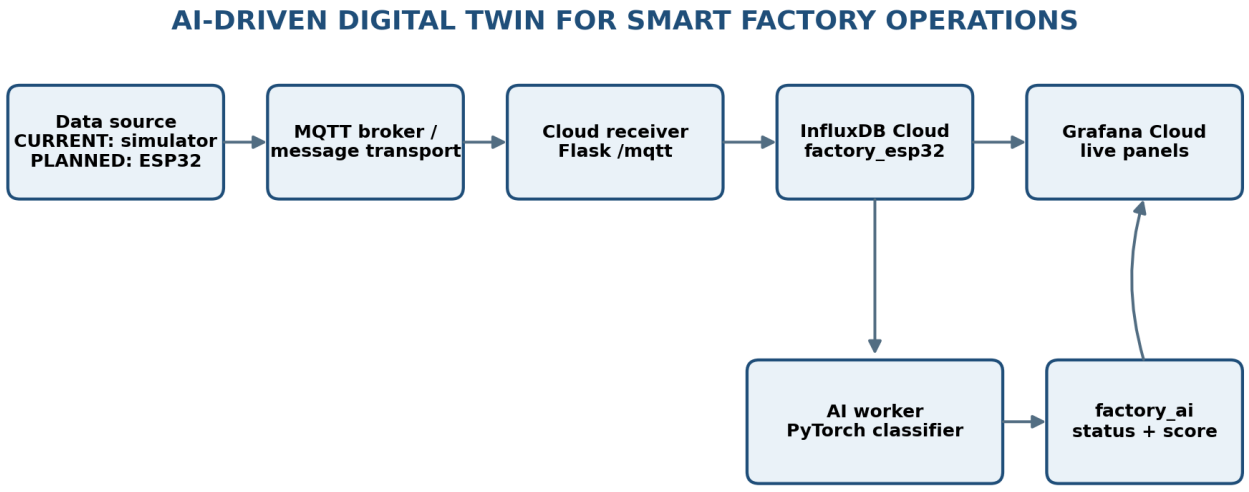
HCLTech Embedded Systems & IoT Project Submission

AI-Driven Digital Twin for Smart Factory Operations

Current status: simulated telemetry prototype; physical ESP32 integration pending firmware review and hardware verification.

Project summary

The prototype receives seven machine parameters, stores telemetry in InfluxDB Cloud, runs a PyTorch classification worker and visualizes machine trends and AI output in Grafana. It should currently be described as a simulated software prototype.



Current prototype uses simulated telemetry. Physical ESP32 acquisition is a planned integration step.

Important implementation facts

• Telemetry fields: temperature, pressure, motor_speed, vibration, current, voltage and humidity.

• InfluxDB measurements: factory_esp32 for telemetry and factory_ai for AI output.

• The supplied cloud AI model uses eight hard-coded training examples and has no independent validation set.

• The receiver currently logs InfluxDB's response but does not check HTTP failure status before returning success.

• The old Grafana query file referenced factory_machine, inconsistent with the receiver's factory_esp32 measurement.

• The supplied repository ZIP does not include the Arduino firmware; the owner reports that it is stored locally.

Submission files

Problem_Statement.docx; SRS_Final.docx; Architecture_Diagram.png; Hardware_BOM_and_Connections.docx; Test_Cases.xlsx; Test_Report_Template.docx; Project_Report_Draft.docx; README.md; grafana/queries.md.

Testing status: Test cases are prepared but have not been executed. Replace Not Run with actual results only after collecting evidence.